

How Much Open Source Is TOO MUCH OPEN SOURCE?

Intel's OpenVINO toolkit helps developers by streamlining code writing, freeing them to concentrate on other vital project aspects. AI Evangelist at Intel, Anisha Udayakumar, elucidates on OpenVINO's versatility



As its name implies, open source knows no bounds and extends even into proprietary hardware companies. Intel's use of open source for model optimisation showcases its ability to nurture a wide array of products in the rapidly expanding field of artificial intelligence and machine learning. This approach has not only showcased the potential of open source but has also established a brilliant business model within a proprietary ecosystem.

With its open visual inference and neural network optimisation (OpenVINO) toolkit, tailored for developers, Intel trains computers to recognise patterns, quickly and accurately. Initially aimed at the computer vision market, this open sourced toolkit has expanded its versatil-

ity to optimise models to understand language or generate content (text or images) through artificial intelligence across various domains, including NLP, generative AI, and emerging large language models.

There is an abundance of open source toolkits in the field of deep learning and neural network optimisation. Whether it's Facebook's PyTorch for research, Keras for user-friendly applications, or Apache's MXNet for unparalleled scalability, the question arises: What sets OpenVINO apart?

With its broad adaptability across various Intel hardware—processors, GPUs, and accelerators—OpenVINO stands out for its specialisation in computer vision and optimisation, making it a strong choice for certain applications. Consider the development of a prototype model for a camera that counts people in a queue, by placing a box around them. The possible delay in inference time can impact retail analytics effectiveness. An effective model with suitable hardware will enhance AI performance by reducing inference time, aiming for real-time results to improve business and operational efficiency. This can be achieved by streamlining the AI model, reducing unnecessary nodes in the deep learning model's architecture, and focusing on essential components.

Developers can choose algorithms that either compromise on accuracy for faster performance or those that offer higher accuracy at the cost of speed, prioritising either based on business requirements.

Developers using the OpenVINO toolkit can run the same code across a spectrum



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